

$$z = \underline{3x^3} + 4x^2y^2 - 6y^3$$

 f_x f_y

$$9x^2 + 8xy^2$$

$$8x^2y - 18y^2$$

 f_{xx} f_{xy} f_{yx} f_{yy}

$$18x + 8y^2$$

$$\underline{\underline{16xy}}$$

$$8x^2 - 36y$$

$$f(x, y) = \underline{\underline{e^{x+y}}}, \quad f_x = e^{x+y}, \quad f_y = e^{x+y},$$

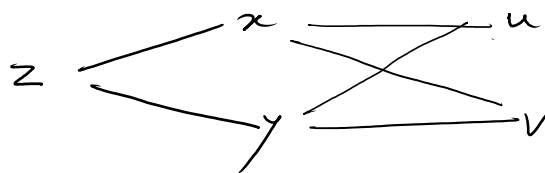
$$f_{xx} = e^{x+y}, \quad f_{yy} = e^{x+y}.$$

$$f_{xy} = e^{x+y}.$$

$$\Delta f = \overset{f}{\left(e^{x+y} \right)} \cdot \Delta x + \left(e^{x+y} \right) (\Delta y) + \frac{1}{2} \left(e^{x+y} \right) (\Delta x)^2 + \frac{1}{2} e^{x+y} (\Delta y)^2$$

$$+ \left(e^{x+y} \right) \cdot \Delta x \cdot \Delta y.$$

$$\simeq f \left(\Delta x + \Delta y + \frac{1}{2} \Delta x^2 + \frac{1}{2} \Delta y^2 + \Delta x \cdot \Delta y \right)$$



$$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy.$$

$$dx = \frac{\partial x}{\partial u} du + \frac{\partial x}{\partial v} dv$$

$$dy = \frac{\partial y}{\partial u} du + \frac{\partial y}{\partial v} dv.$$

$$\begin{aligned} dz &= \frac{\partial z}{\partial x} \left(\frac{\partial x}{\partial u} du + \frac{\partial x}{\partial v} dv \right) + \frac{\partial z}{\partial y} \left(\frac{\partial y}{\partial u} du + \frac{\partial y}{\partial v} dv \right) \\ &= \left(\frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial u} \right) du + \left(\frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial v} \right) dv \end{aligned}$$

$$W = x^2 + y^2$$

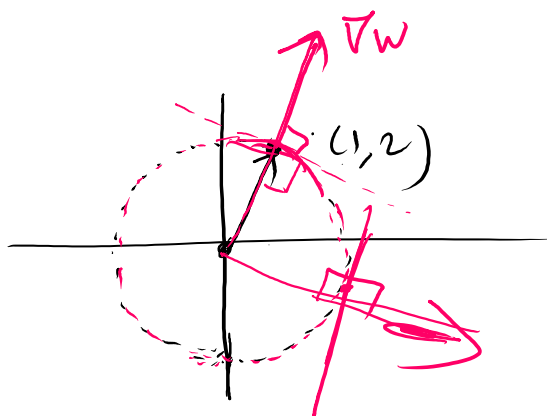
$$x^2 + y^2 = c$$

$$c = 1, 2, 3, 0.5$$

$$x_0 = 1, y_0 = 2$$

$$1^2 + 2^2 = c = 5$$

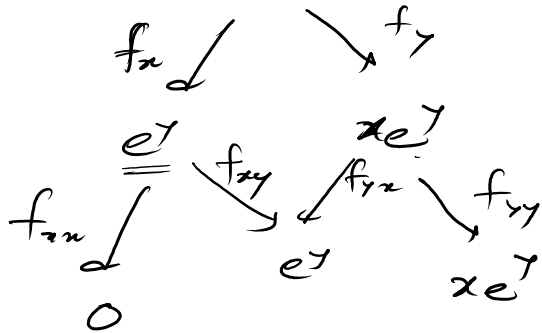
$$x^2 + y^2 = 5$$



$$\nabla W \Big|_{\substack{x_0=1, \\ y_0=2}} = \begin{pmatrix} 2x \\ 2y \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$$

$$f(x, y) = xe^y,$$

$$\underline{x_0=0, y_0=0}, \quad \mathbf{x}_0 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$



$$f(x+\Delta x, y+\Delta y)$$

$$= f(x, y) + f_x \Delta x + f_y \Delta y + \frac{1}{2} f_{xx} (\Delta x)^2 + \frac{1}{2} f_{yy} (\Delta y)^2 + f_{xy} (\Delta x) (\Delta y)$$

$$= f(x, y) + e^y \Delta x + xe^y \Delta y + \frac{1}{2} \times 0 \times (\Delta x)^2 + \frac{1}{2} xe^y (\Delta y)^2 + e^y \cdot (\Delta x) (\Delta y)$$

$$= 0 + \Delta x + 0 + 0 + 0 + \Delta x \cdot \Delta y$$

$$= \Delta x + \Delta x \cdot \Delta y$$

$$f(x) = f(x_0) + \underline{\nabla f(x_0)} \cdot (x - x_0) + \frac{1}{2} (x - x_0)^T \cdot \underline{H_f(x_0)} \cdot (x - x_0)$$

$$x_0 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad x - x_0 = \begin{pmatrix} \Delta x \\ \Delta y \end{pmatrix}$$

$$\underline{\nabla f} = \begin{pmatrix} \underline{e^y} \\ \underline{x e^x} \end{pmatrix}, \quad H = \begin{pmatrix} 0 & e^y \\ e^y & x e^x \end{pmatrix}$$

$$\nabla f|_{x=x_0} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad H = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\begin{aligned} f(x) &= f(x_0) + \begin{pmatrix} 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} \Delta x \\ \Delta y \end{pmatrix} + \frac{1}{2} \begin{pmatrix} \Delta x & \Delta y \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \Delta x \\ \Delta y \end{pmatrix} \\ &= 0 + \Delta x + \frac{1}{2} \begin{pmatrix} \underline{\Delta x} & \underline{\Delta y} \end{pmatrix} \begin{pmatrix} \underline{\Delta y} \\ \underline{\Delta x} \end{pmatrix} \\ &= \Delta x + \Delta x \cdot \Delta y \end{aligned}$$